

SUMUKH VARMA Y

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PROFESSIONAL SUMMARY

Backend-focused software engineer with 3 years of experience building and scaling distributed systems in cloud-native environments. Specializes in Node.js, TypeScript, PostgreSQL, Python and AWS. Known for systematic incident investigation, database performance optimization, and driving measurable reliability improvements in production systems and taking on end-to-end ownership.

TECHNICAL SKILLS

Languages & Frameworks: Node.js, ReactJs, TypeScript, Python, Django, FastAPI, SQL, NoSQL

Cloud & Infrastructure: AWS (SQS, SNS, Lambda, SES, S3, API Gateway, RDS, EKS, ECR), Kubernetes, Docker

Databases & Tools: PostgreSQL, MongoDB, DynamoDB, Redis, TypeORM, Git, Keycloak, MCP, Agentic frameworks, Grafana telemetry

WORK EXPERIENCE

Software Engineer | IHX (A Perfios Company)

Sep 2024 – Present

Orchestration Engine • Persistence Engine • Acclaim Backend • Infrastructure

- Systems process 20,000 insurance claims per day with a failure rate of ~2 claims/day — 99.99% reliability
- Architected dynamic claim processing pipelines with runtime adaptation and context-based fallback routing; introduced Redis-based sliding-window rate limiting to prevent overload during peak traffic
- Built pipeline timing controls enabling safe production validation of new pipelines on small traffic subsets before full rollout
- Introduced Dead Letter Queues and retry logic across claim creation and update flows — reduced OPS incidents requiring manual intervention by eliminating full queue blockages caused by single claim failures
- Replaced ~100 individual DB writes per claim with a single bulk persistence API call — cut write operations by ~99% per claim, reducing DB connection usage and latency across all services
- Diagnosed and resolved recurring production timeouts on the persistence service: the database query planner was working with stale statistics, causing it to scan 646,000 rows to return 179 results on every dashboard load. Applied targeted indexes and database maintenance — reduced average RDS CPU utilization from ~80% to sub-30%, and cut query execution time by over 95%
- Optimized Report Summary API from 2+ minutes (or crashes) to a consistent 10–15 seconds for 3-month ranges using streaming and parallelization — an 87%+ improvement in response time
- Delivered PAE integration end-to-end with stable go-live and no regressions; migrated vendor allocation to SQS standard queues, eliminating cascading failures where one blocked claim stalled 200–300 others
- Introduced agentic AI workflows within the engineering team, improving developer productivity and reducing time spent on repetitive investigation and boilerplate tasks
- Set up Keycloak Email MFA for HDFC-ERGO: configured browser flows, OTP themes, and wrote replication documentation for future deployments
- Built RESTful APIs for healthcare document digitization services leveraging FastAPI's asynchronous capabilities with Uvicorn/Gunicorn ASGI servers for high-performance request handling; integrated Azure AI Form Recognizer and MongoDB for intelligent document processing workflows
- Deployed Python microservices with production-grade features including graceful shutdown handling, CORS middleware, environment-based configuration, Grafana telemetry

On the frontend — Identified that onboarding each new insurance client required duplicating ~29,000 lines of hand-written JSON config across 3 screens. Designed and built a V3 config-driven module system (69 TypeScript modules, transformer layer, registry pattern) that reduces per-client config to ~500 lines — a **98% reduction** — while preserving full backward compatibility with the existing React + Redux render pipeline. Delivered end-to-end across 20 phases with 780+ automated tests at 96%+ coverage.

- Implemented background fire-and-forget API calls for long-running operations, improving UI responsiveness for end users

Software Engineer | Mobil80 Solutions

Jul 2023 – Aug 2024

HelloWrk • Posture.ai • StepSync

- Led backend engineering for HelloWrk CRM — integrated SIP protocol for third-party call providers and built a full email client using AWS SES

- Architected serverless infrastructure using AWS Lambda, API Gateway, AppSync, SQS, DynamoDB, and EventBridge for multiple SaaS products
- Built AI/ML posture analysis platform (Posture.ai) with OpenID integrations and automated health-concern detection algorithms
- Led StepSync fitness app backend — resolved cross-platform fitness tracker compatibility and built centralized metrics tracking across watchOS devices
- Directed teams of 4–5 engineers; handled client discussions, stakeholder requirements, and end-to-end delivery

Software Trainee | Bosch Global Software Technologies

Jan – May 2023

- Built a fully autonomous maze-solving robot using Arduino and C++; engineered a roll-over detection system with real-time tilt alerts

Engineering Trainee | Bharat Electronics Ltd

Sep – Oct 2022

- Trained on end-to-end product development for Advanced Naval Systems including CNC machining, ERP/SAP, and quality assurance

EDUCATION

B.E., Ramaiah Institute of Technology, Bangalore

Aug 2019 – Aug 2023 | GPA: 8.47 / 10.0

- Published research paper: Modelling and Control of Tendon Driven Continuum Robot — IJNRD, Vol. 8, Issue 8, Aug 2023 (Impact Factor: 8.76)

CAREER FOCUS & GROWTH AREAS

- Pursuing architectural ownership — system design decisions that impact scalability across services
- Growing into mentorship and cross-team technical leadership; building deeper knowledge of service interactions
- Expanding to dev tooling, infrastructure complexity, and larger collaborative initiatives

ACHIEVEMENTS

- 10+ Golds at inter-collegiate basketball; 2 Golds at University Games; State-level athletics competitor representing Karnataka nationally (2017–2019)
- Sports Captain, Sindhi High School — led 12-member student council
- Volunteered with MARSHALL INDIA 2021 under Swachh Bharat initiatives, collaborating with the Government of Karnataka